TEXAS A&M UNIVERSITY
RED
ROCKET ENGINE DESIGN

Design Concept Review

LOx Engine

- TCA
- FSYS
- AVI
- STR
- SFE

Contributors and Responsible Engineers

Chief Engineer: Michael Guarnery

TCA

- Leads: Nick Mancini, Nicolas Adame
- Regen Nozzle: Jaxon Bain, Matthew Haban
- Injector: Sharath Kumar
- Ignitor: Ella Gomez, Michael Sesate

FSYS

- Leads: Garrett Adams, Garrett Swail (E2)
- LOx: Steven Mendelsohn
- Jet-A: Merrick Estes
- GN2/Pneumatics: Josh Seward
- Ignitor: Nate Seward

STR

- Leads: Rishabh Menon, Brandon Laws
- Support Structures: Jaden Smith

AVI

- Leads: Joshua Mark, Juniper Mills
- AVI Integration: Ryo Kato

SAFE

- Leads: Riley Reid, Logan Williams
- TCA: Michael Sesate
- FSYS: Logan W.
- STR: Quentin C. & Thomas G.

Ragnarok Requirements High-Level

- Ragnarok shall produce 2000 lbf of thrust
  - Driven by nozzle geometry
- Ragnarok shall use LOx and Jet-A
- Ragnarok shall be regeneratively cooled
- Ragnarok shall be remotely operated
- Ragnarok TCA shall be designed to 1.5 FOS on yield and 2.0 FOS on ultimate stress
- Ragnarok tanks shall be designed to 1.75 FOS on yield and 3.0 FOS on ultimate
- Ragnarok structures shall be designed to 4.0 FOS on yield

TCA

I. Requirements
II. Materials Trade Study
III. Injector Trade Study
IV. Torch Igniter
V. Thrust Chamber
VI. Methods for sizing and analysis

TCA Requirements

- Ragnarok TCA shall produce 2000lbf of thrust for 10 seconds
- Ragnarok TCA shall be regeneratively cooled
- Ragnarok TCA shall have a chamber pressure above 450 psi
- Ragnarok TCA shall be manufactured using additive manufacturing
- Ragnarok TCA shall be be capable of rapid restart
- Ragnarok TCA shall maintain an O/F ratio between 2.0 and 2.4

TCA Architecture and Overview

TCA Dimension Estimates

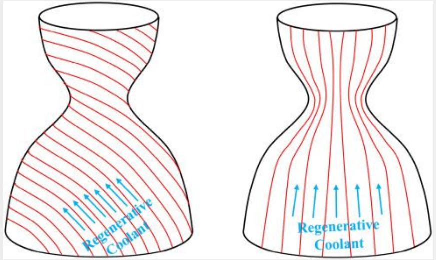
Preliminary design estimates Jan 3, 2025

Chamber
- OF ratio: OF = 2.22 - 2.29 (final OF will depend on final dimensions)
    - OF ratio for maximum ISP assuming frozen composition at the throat (more accurate than equilibrium assumption)
- Pc = 500 psi
- mdot = 3.32 kg/s
    - For our 9 kN thrust goal and a known expansion ratio and Pc, this mdot is required
- Characteristic length: L* = 1.20
    - Middle of the range for RP1/LOX engines (1.0 - 1.5), longer will provide more complete combustion but heavier
- Contraction ratio: Ac/At = 6 - 10
    - Higher CR allows for more time for vaporization
- Chamber area: Ac = 10.5 E-3 m^2 to 17.5 E-3 m^2 (depending on chosen CR)

Nozzle
- Throat size: At = 1.75 E-3 m^2 or dt = 4.686 cm
    - This provides the mass flow rate required for thrust goal, for given pressure and optimal expansion
- Exit size: Ae = 10.15 E-3 m^2
    - Based on expansion ratio
- Expansion ratio: Ae/At ~5.5 (RPA) - 5.8 (hand calc w/ CEA)
    - Expansion ratio that gives optimal expansion at sea level, much larger may produce flow separation
- Nozzle type: Full length conical nozzle (just to get a max nozzle length)
    - rt = 0.0234 m
    - re = 0.0568 m
    - Conical half-angle: 15 degrees
    - L cone = 0.1240 m

Total Dimensions
- Max length: 33.1 cm
    - 20.7 cm from injector face to throat
    - 12.4 cm from throat to exit
- Max diameter: 11.4 cm (chamber diameter)

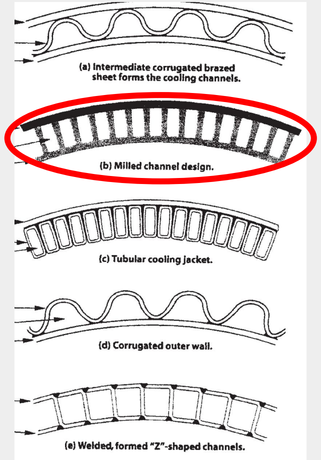
Cooling Designs



- RPE estimates that axial cooling channels will be ineffective for flow rates < 9 kg/s (swirled channels likely needed)

considerations. *Axial flow cooling jackets*, or *tubular walls*, have low hydraulic friction losses but are practical only with large coolant flows (above approximately 9 kg/sec); for small coolant flows and small thrust units, design tolerances of the cooling jacket width between the inner and outer walls or the diameters of the tubes become too small, or dimensional tolerances become prohibitive. Hence, most small thrust chambers use radiation cooling or ablative materials.

Rocket Propulsion Elements, 9th Edition, P. 291



- JetA fuel will be used as coolant before entering combustion chamber (**mdot ~ 1.08 kg/s**)
- Injection near the nozzle exit gives simplest design
- "Milled" channel shape offers the greatest heat transfer and is the obvious choice for channel shape (especially for 3D printed chamber)

Injector Trade Study

- Impinging element candidate: unlike split triplet (tldr: easiest to manufacture, lowest L*, best wall compatibility, combustion instability risk requires extra design consideration, largest literature base to inform/guide design)
  - Impingement type injectors achieve higher characteristic velocity, combustion efficiency
  - The unlike impinging doublet and triplet are the most popular element types in small engines where performance requires rapid combustion; large literature base
  - O-F-O triplet configurations yield highest performance, but have poor chamber wall compatibility, higher risk of hardware failure
    - Common mod is to add outermost injector element ring of showerhead type, flowing fuel to film cool, or instead use F-O-F configuration
  - The unlike triplet allows very rapid mixing and atomization, minimizes L* and thus cooling demand of regen system
  - Impinging elements offer best wall compatibility with control of droplet size distribution; increased droplet size near wall, by increasing the diameter of peripheral injector face orifices, provides a stabilizing effect in the region of highest chamber pressure oscillations (the wall)
  - Higher performance is achieved at larger impingement angles (60-deg or higher), increasing injector face thermal load
  - Easiest geometry to manufacture: sensitive to manufacturing tolerances, but requires simpler tooling/machinery
  - Most susceptible to combustion instabilities, requires intentional design
  - Well suited for throttling, very stable (simpler manifold geometries)
  - With heavier hydrocarbon fuels, lower inherent volatility must be compensated by vaporization performance of injector
  - Triplets do not need a large number of very small holes to satisfy vaporization requirements for good performance
  - F-O-O-F split triplet is less sensitive to changes in momentum ratio than the parent F-O-F triplet, allowing better performance during transients (for the intended 10-second burn of Rag, a significant portion is at transient conditions)

Injector Trade Study

- Non-impinging element candidate: swirl bi-coaxial (tldr: hardest to manufacture, low L*, smallest literature base, lower risk of combustion instability but largely thought to be for gas/liquid coaxs)
  - Highest manufacturing complexity
  - Rapid mixing and atomization, lowers L* and thus cooling demand of regen system
  - Most design correlations study gas/liquid injectants, with oxygen/hydrogen propellant combinations
    - Limited literature to reference liquid/liquid LOX/Jet-A injectant designs
    - YJSP used an open-type swirl coax (switched from pintle) for their LOX/Jet-A, 2500lbf flight engine (design reference)
  - Higher stability of coax is primarily attributed to different injectant phases, rather than element type
    - Stability benefits, with a liquid/liquid swirl bi-coax, over the unlike split triplet may be negligible
  - The required high velocity ratios and lower limit of fuel-to-oxidizer area ratios limits most applications to gas/liquid

Injector Trade Study

- Hybrid element candidate: pintle (simple manufacturing for static pintle, more complex for dynamic pintle, lowest risk of combustion instability, highest L*)
    - Relatively simple manufacturing without throttling (no actuated sleeve)
    - Best geometry for stability, attributed to mantle and core recirculation zones
    - Not as rapid mixing/atomization, forces a longer L* (more weight for flight vehicle, higher cooling demand on regen system)
    - Often used in single-element injector assemblies; if Jet-A is flowed through annular gap, efficiently sized pintle features the largest orifice diameters to minimize FOD and coking related issues
    - Mantle recirculation zones offer cooling of injector face and injector adjacent walls if fuel is annular propellant

Injector Materials Trade Study

https://docs.google.com/spreadsheets/d/1xmGCG89nT8Og2eIuagca2__nxfY74-fmIrX9He6VofA/edit?gid=0#gid=0

- SS304/SS316 are leading candidates over copper, aluminum, titanium alloys
  - Cryogenic LOX compatibility
  - Highest yield strength at temperatures predicted during fire (95 MPa at 700 C for SS304)
    - SS304 used by RED on Elysium 2 injector (150 psi higher chamber pressure)
  - Differ in CTE from likely chamber copper alloy GRCop-42 by ~1 ppm/K at these temperatures
  - Low thermal conductivity at fire temperatures (.149 W/mK at 500 C for SS304)
    - Minimizes heat transfer to propellants before injection
    - Prop densities more closely approximated by inlet temps
    - Accurate orifice sizings, momentum ratios of propellants easier to calculate
  - 1/3 price of C18150 per volume

Nozzle & Thrust Chamber Materials Trade Study

- Ruled out non-copper alloys due to substantial lack of thermal conductivity
  - Ex. ~300 W/mK for copper alloys vs ~20 W/mK for stainless
- Focus on 4 copper alloys:
  - CuNi2SiCr (High availability)
    - Strength at elevated temperature is mostly described qualitatively and inconsistently, but is said to be somewhat similar to CuCrZr (not ideal)
  - CuCrZr (Somewhat available)
    - Recommended temperature limit is 350 C for “long-term operation,” but maintains strength of 80-90 MPa at 700 C.
  - GRcop-84 (Extremely low availability)
    - Not recommended due to lack of development compared to superior -42 variant
  - GRcop-42 (Low availability)
    - Top choice due to superior thermal conductivity (344 W/mK), retention of strength at 700 C (>105 MPa), creep resistance, and its intentional design case for combustion chambers.

https://docs.google.com/spreadsheets/d/1PTn6ZJLD0P5bb_E0aHCncmRX2k5OEh2-xSR31MHD4DI/edit?usp=sharing

Ignitor Injector Type Trade Study

https://docs.google.com/document/d/18WuR9auLpSjZLT3b6Ggw1cPqtdXLPXWy_GnzwZisnVc/edit?usp=sharing

- Flow rates currently around .055 kg/s GOX, .035 kg/s Jet-A
- Single unlike doublet element chosen for first prototype
    - FOD contamination concerns minimized for reliable operation
        - At pressure drop, both orifices around 3/32” diameter
    - Mixing comparable to other, more complicated designs
    - Will be evaluated during Jet-A - Nitrox testing
- Other designs
    - F-O-F triplet
        - Simple but would require 3/64” orifices for Jet-A
        - More predictable performance
    - Coaxial/Pintle designs
        - Even smaller orifices/gaps for both fuel and oxidizer
        - O-ring + two piece injector construction
        - Much more complicated

Ignition Method Trade Study

https://docs.google.com/document/d/11g5FucUdRTFheCwC4bS2ycPJp84LIVaVu06-LxoJZRc/edit?usp=sharing

- Car spark plugs preferred for balance between durability and compactness (thread to electrode ~4mm)
- Other options
  - Glow plugs had thread to electrode distance of around 30-40 mm
    - limit placement
    - require lots of stock and space on TCA
  - RC spark plugs smaller than full size but
    - designed for 200 psi RC engines
    - have only found success in ~200 psi amateur torch igniters
    - Our igniter will target 500-600 psi pressure
  - RC diesel plugs almost always one use in amateur torch igniters

Ignitor Spark Plug Trade Study

https://docs.google.com/document/d/11Rfgrg6LHRxe_t2tUNj3DAgahl1wJSC3RUY3hzYDtkU/edit?usp=sharing

- Material of electrode tip and electrode core must be iridium
  - Melting point of iridium: 2450 C, slightly lower than flame temperature
  - Torch will only be running for ~2 seconds so steady max flame temperature will not be reached
- NGK Iridium IX or Bosch Double Iridium
  - Both are made for high performance vehicles and have iridium electrodes
  - Cost 10-15$ per plug

TCA 3D Printing Companies

Can print GRcop-42

- I3DMFG
- 3DSystems
- Velo3D

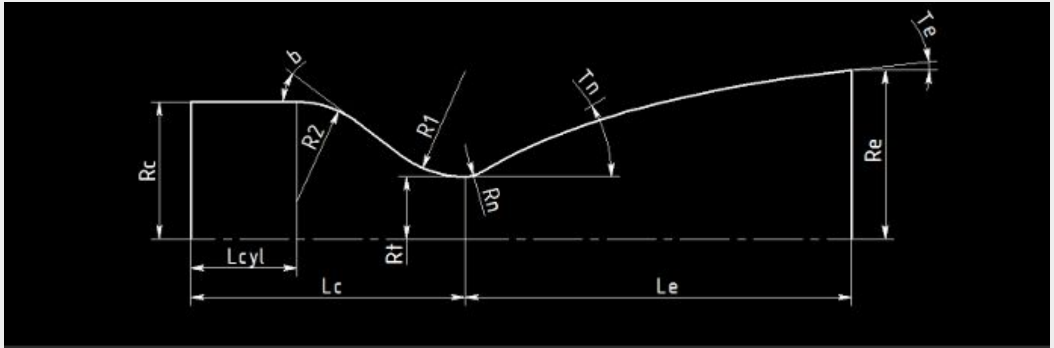
Can’t print GRcop-42

- Xometry
- CraftCloud
- Protolabs
- JLCPCB
- HLH

Methods of Sizing and Analysis

- RPA: Combustion properties, sizing nozzle, preliminary regen cooling sizing
- Ansys: Structural, thermal, and CFD analysis
  - Grace Supercomputer at A&M for Ansys
- Injector sizing code?
  - We have one for pintle that can be modified
- Regen cooling sizing code:
  - The code implements a detailed physics based 1-D nozzle and regenerative cooling framework, combining CEA derived freestream properties with isentropic flow modeling, iterative gas side convective heat transfer, wall conduction using temperature dependent material models, coolant side convection driven by explicit channel geometry and fin efficiency effects through Gnielinski based correlations, with axial coolant energy marching to account for coupled thermal behavior along the nozzle.
  - The gas side wall temperatures predicted by the code show good agreement with the preliminary Ansys simulations..

RPA - Chamber and Nozzle Sizing

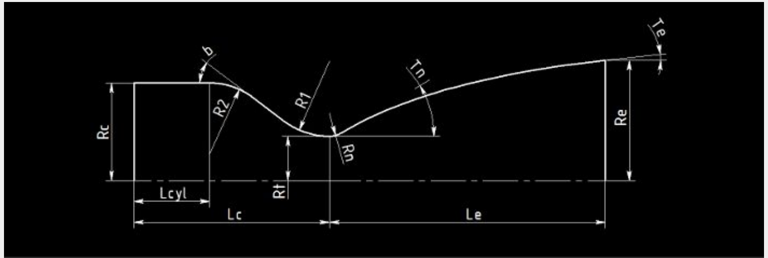


Combustion chamber size		
Dc	133.93	mm
Dt	47.35	mm
Lcyl	105.18	mm
Lc	186.44	mm
L*	1200.00	mm
R1	35.51	mm
R2	56.14	mm
b	45.00	deg
Ac/At	8.00	

Nozzle size		
Type	Parabolic nozzle	
Rn	9.04	mm
Tn	20.52	deg
Te	8.00	deg
De	111.21	mm
Le	128.78	mm
Le/Dt	2.72	
Le/Lc15	107.01	%
Ae/At	5.52	

- Expansion ratio larger than ~5.5 - 5.8 is overexpanded and may risk flow separation
- This sets mass flow to ~3.4 kg/s, and throat diameter to 4.6 cm - **matches hand calcs**

Parameterized model



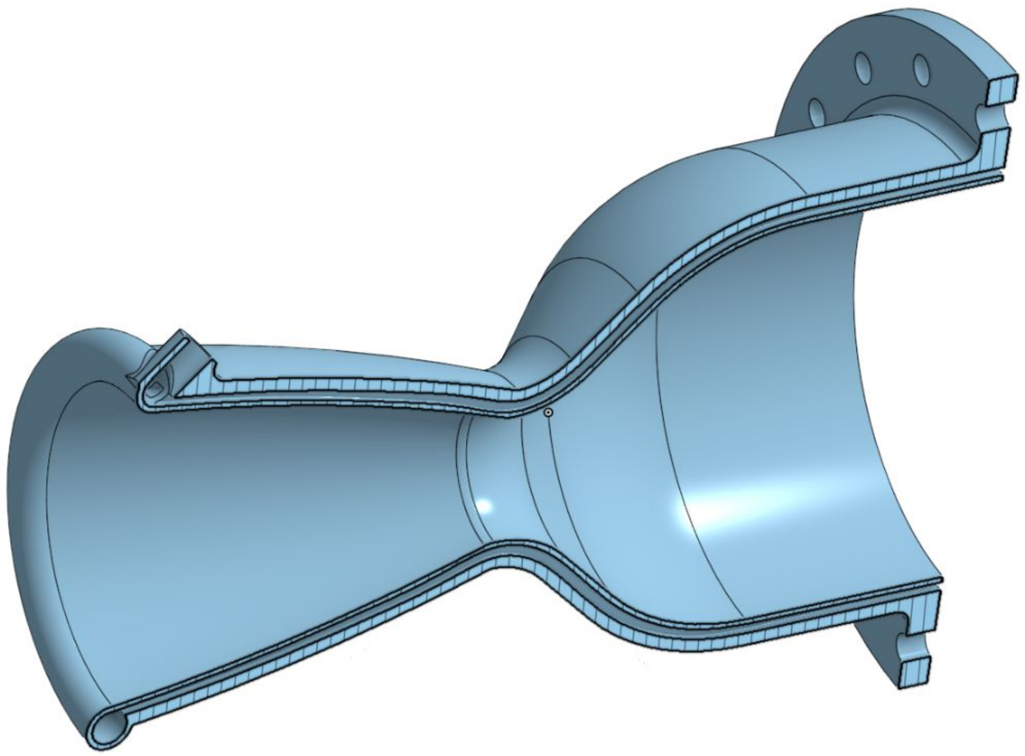
Combustion chamber size

Dc	133.93	mm
Dt	47.35	mm
Lcyl	97.89	mm
Lc	189.62	mm
L*	1200.00	mm
R1	35.51	mm
R2	74.76	mm
b	40.00	deg
Ac/At	8.00	

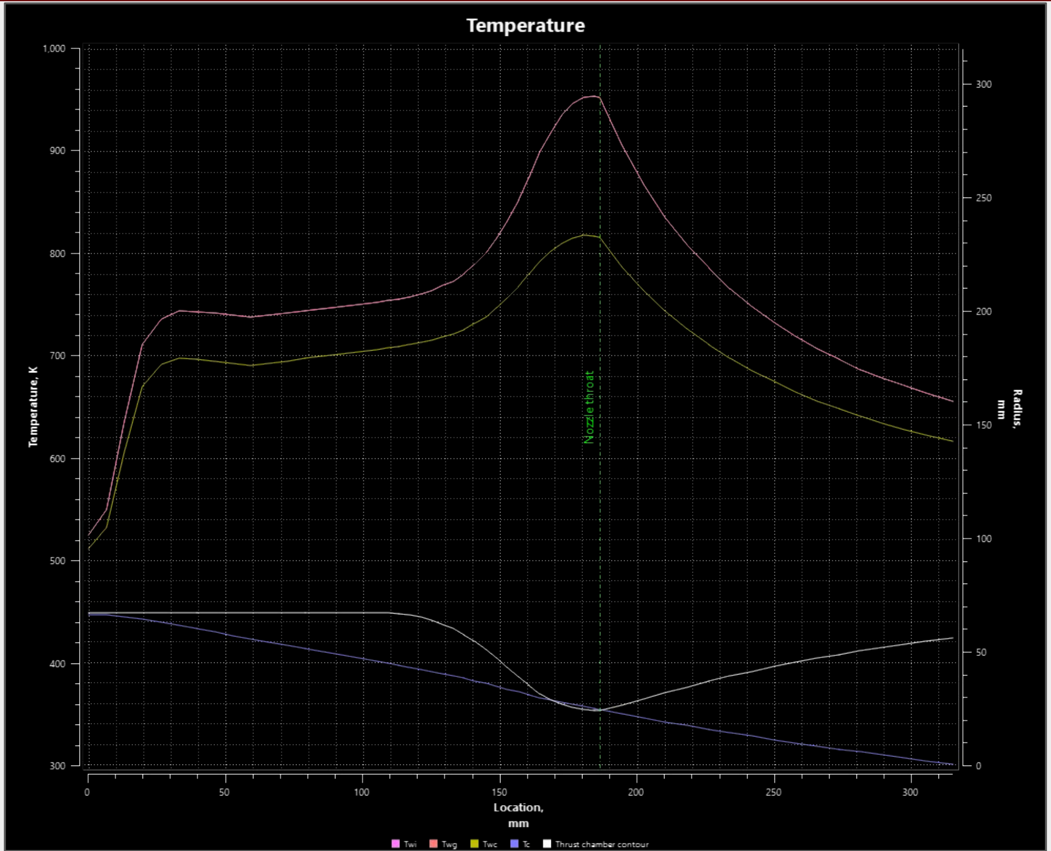
Nozzle size

Type	Parabolic nozzle	
Rn	9.04	mm
Tn	20.52	deg
Te	8.00	deg
De	111.21	mm
Le	128.78	mm
Le/Dt	2.72	
Le/Lc15	107.01	%
Ae/At	5.52	

Parameter		**Engine**	**Chamber**	
Thrust	sea level	8.8964	8.8964	kN
	opt exp	8.9093	8.9093	kN
	vacuum	9.8806	9.8806	kN
Specific Impulse	sea level	2575.9759	2575.9759	N-s/kg
	opt exp	2579.7022	2579.7022	N-s/kg
	vacuun	2860.9341	2860.9341	N-s/kg
Mass flow rate	total	3.4536	3.4536	kg/s
	oxidizer	2.3744	2.3744	kg/s
	fuel	1.0793	1.0793	kg/s
	film coolant	n.a.	n.a.	kg/s
Number of chambers		1		



RPA - First Pass Thermal Analysis



Ansys Analysis Pipeline

- Designs will change often due to iteration
- Need a standard analysis pipeline to accurately compare iterations and simplify the process of performing analysis
- 2D CFD nozzle and chamber with post combustion speciation
- 3D CFD Regen Channels with heat load from RPA or 2D Ansys nozzle CFD
  - Find film coefficient along regen changes and temp of regen coolant
  - Find dP and velocity through regen channels
- Steady state and transient thermal analysis
  - Steady state is worst case and transient will simulate startup
- Steady state structural and modal analysis
  - Import thermal load and add pressure loads

Estimating Film Coefficient

Channel Geometry

- 2mm x 2mm cross section
- Area A=4e-6 m^2
- Perimeter: P=8e-3 m
- Roughness R=0.05 mm, Dh=4*A/P0.002

Assumptions

- Velocity V=3 m/s
- Length L=0.3m
- Bulk mean temp: 400k
- Viscosity: $\mathcal{V}$=4.6e-7 m^2/s (JP-8 @ 400k)
- Density: ρ=740 kg/m^3 (@400k)
- Dynamic viscosity: $\mu=\rho\mathcal{V}$=3.4e-4 N-s/m^2
- Specific heat: Cp=2340 J/kg-k
- Conductivity: K=0.101 w/m-k
- Prandtl Number: Pr=Cp*μ/K = 7.9
- Reynolds Number=ρ*V*Dh/μ = 1.31e4
- Friction factor: f=0.056
- Nusselt Number: Nu=161
- Film coefficient: h=Nu*K/Dh = 8000 Wm^2/K
- Error: ±30 %, h=5000 Wm^2/K very conservative estimate

Correlation: Gnielinski

$$Nu_{Dh} = \frac{(f/8)(Re_{Dh} - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)}$$

where:
Dh is the hydraulic diameter [m]
Re is the Reynolds number [-]
Pr is the Prandtl number [-]
Nu is the Nusselt number [-]
f is the Darcy friction factor [-]

Validity:

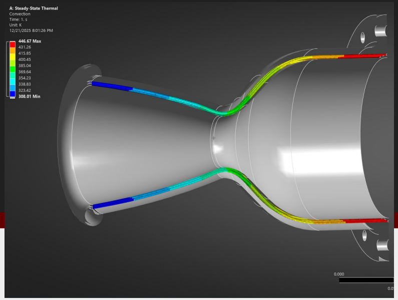
$$0.5 \le Pr \le 2000$$

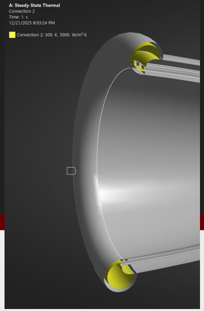
$$3000 \le Re_{Dh} \le 5 \times 10^6$$

$$f = \frac{1.325}{\left[\ln\left(\frac{\varepsilon}{3.7D} + \frac{5.74}{R_e^{0.9}}\right)\right]^2}$$

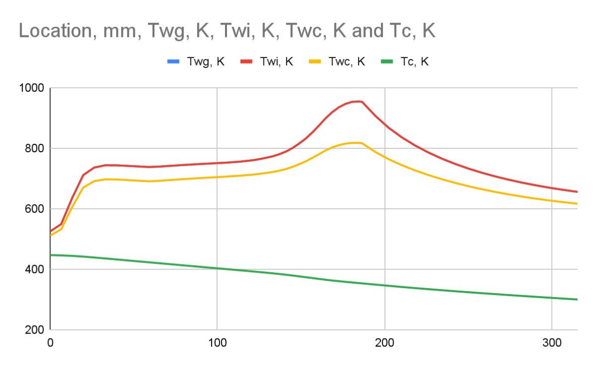
Where			
	f	=	Friction factor (unitless)
	ε	=	Roughness height (m, ft.)
	D	=	Pipe diameter (m, ft.)
	R_e	=	Reynolds Number (unitless)

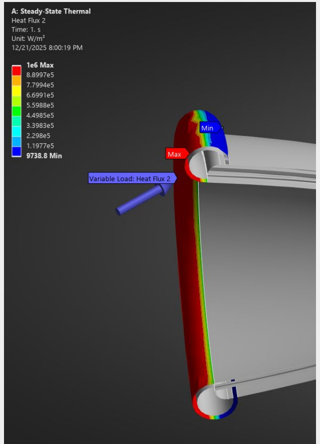
Ansys - Thermal Analysis Loads

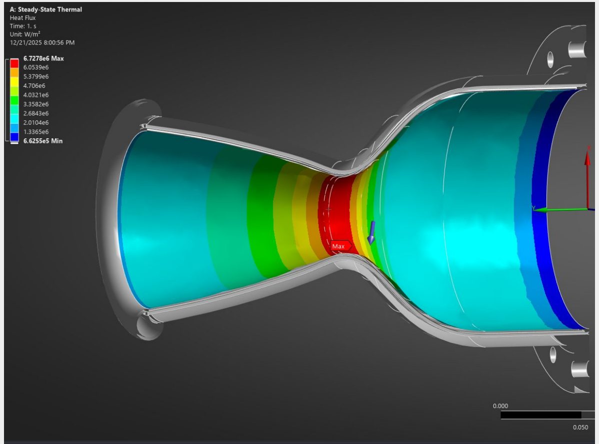




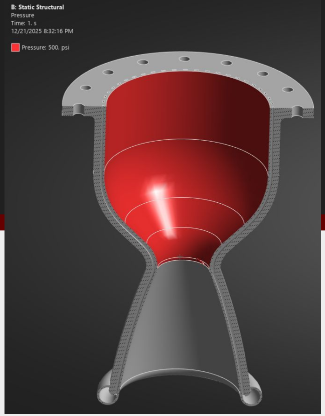
- Convection
  - Fuel annulus input: h=8000 W/m^2-k, T=300 k
  - Regen channels: h=8000 W/m^2-k, T= RPA Tc (coolant temperature gradient)
  - External surfaces: h=30 W/m^2-k, T=300k (due to natural convection from environment)
- Heat flux:
  - RPA Total Heat Flux (includes convection + radiation inside chamber/nozzle)
  - Fuel annulus heat flux q= 1e6 W/m^2 (from flame interactions with fuel annulus)
- Radiation: Outside surfaces

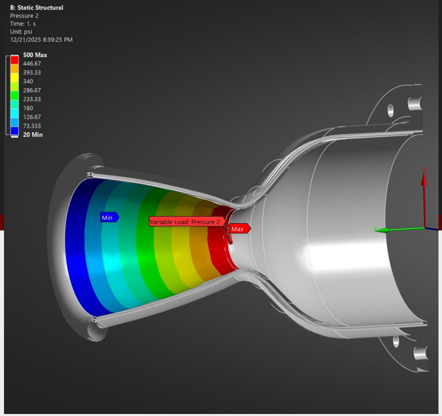






Ansys - Structural Analysis Loads



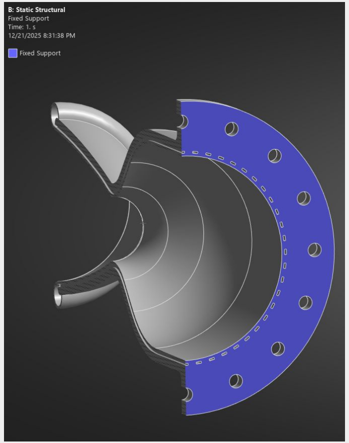


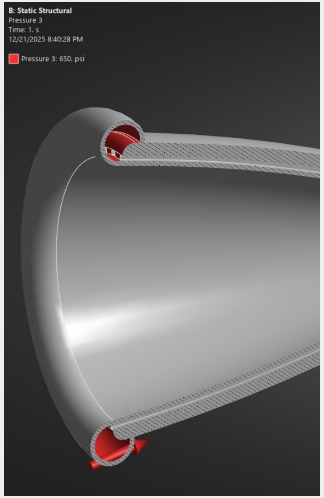
Pressure Loads:

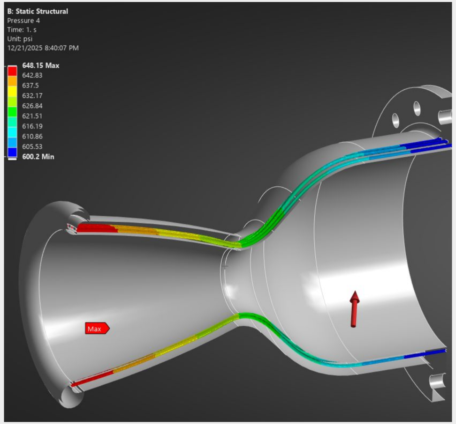
- Chamber: 500 psi
- Nozzle: 500 psi (throat) to 20 psi (nozzle)
- Regen channels: 650 psi (nozzle) to 600 psi (injector)
- Fuel annulus/inlet: 650 psi

Fixed Support:

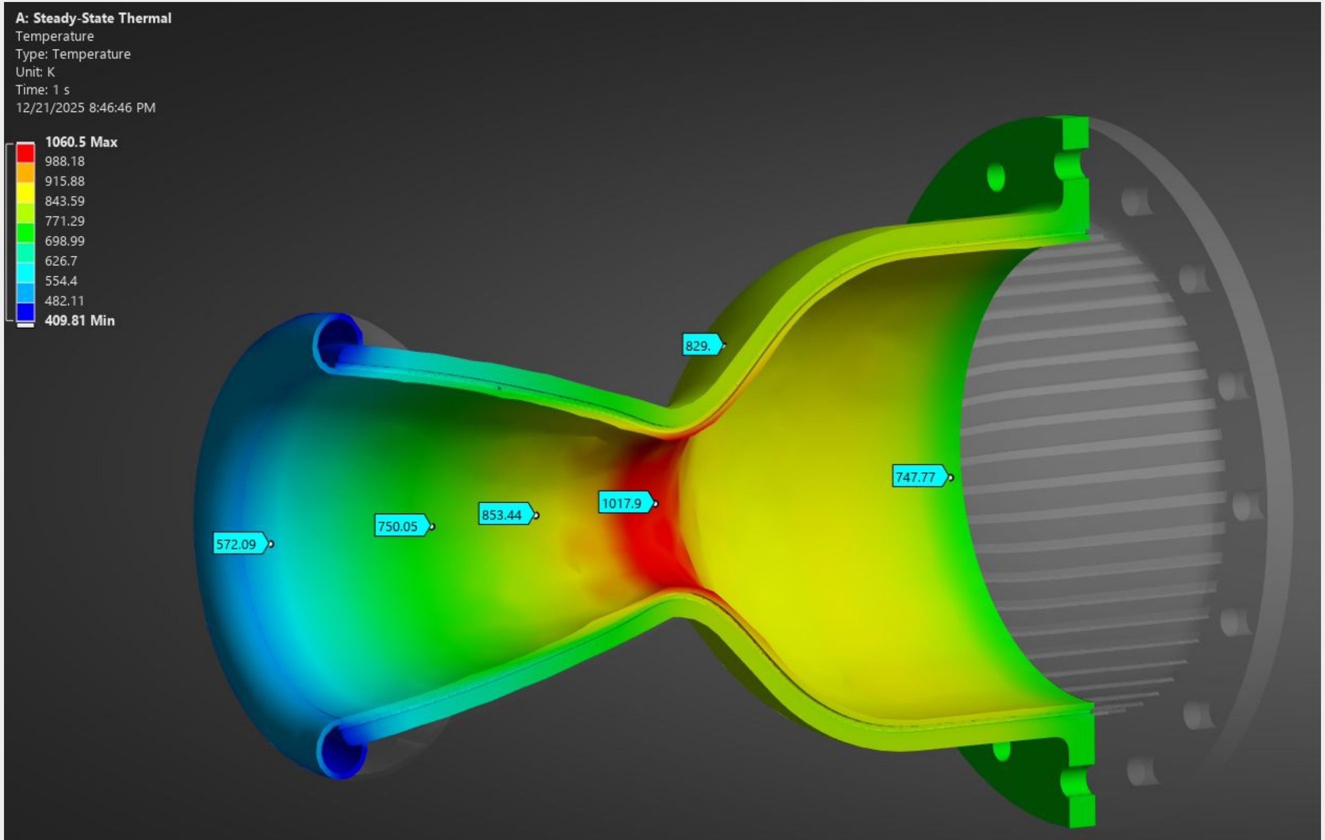
- Back of flange



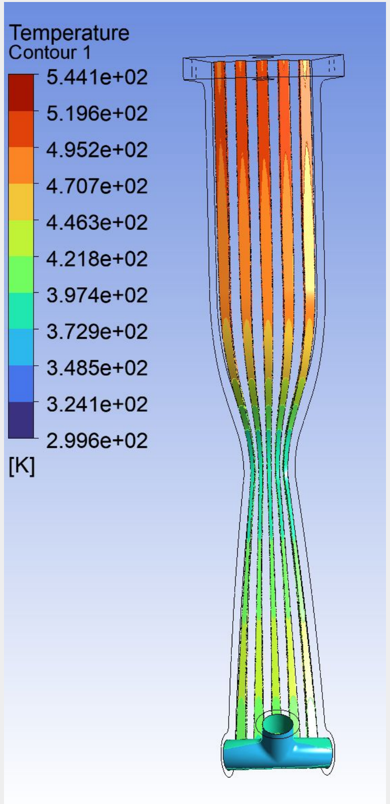


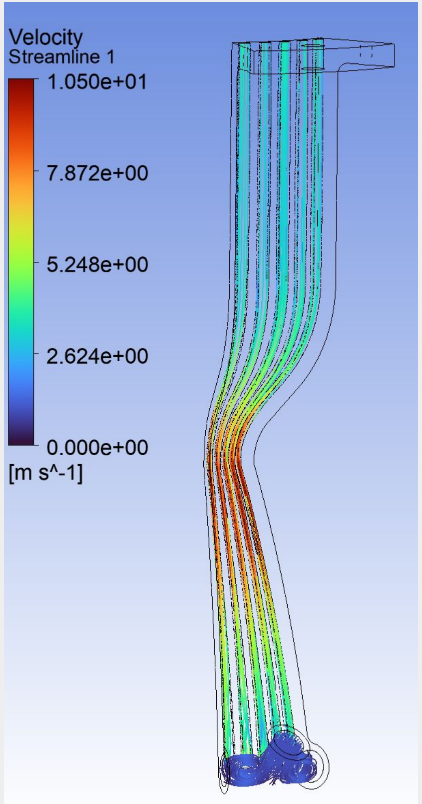


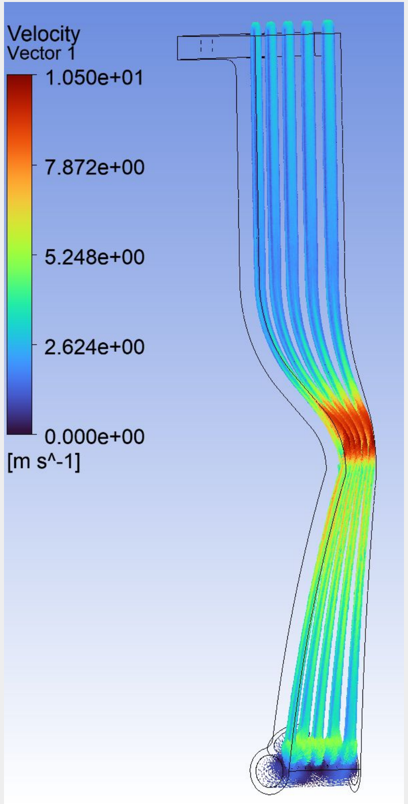
Ansys - Steady State Thermal Analysis

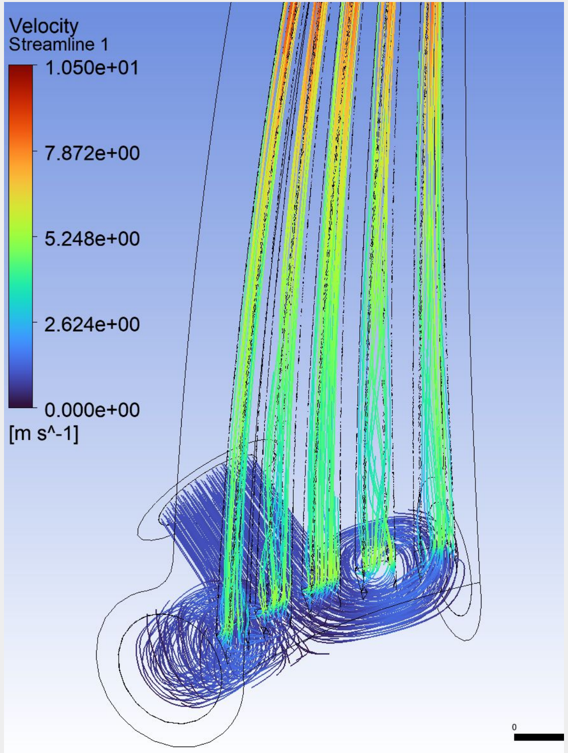


Ansys - Regen Channel CFD and Thermals









FSYS

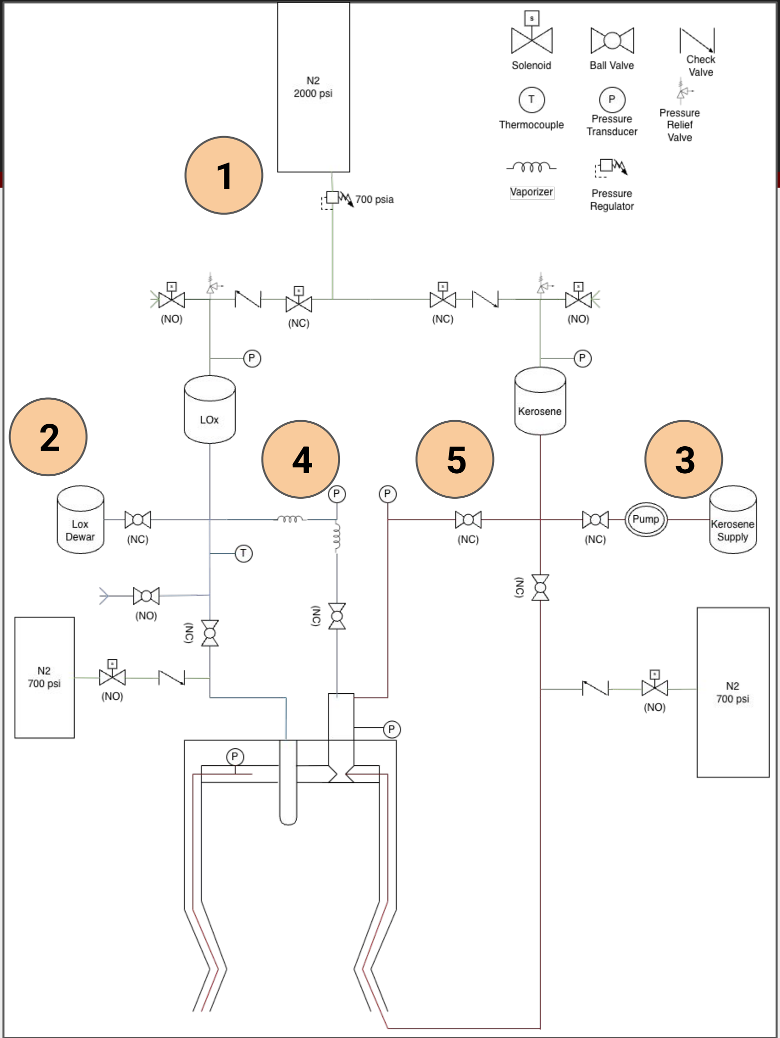
I. Requirements
II. P&IDs
III. CONOPS
IV. Tanks
V. LOX
   a. Valve Modifications
VI. Jet-A
VII. GN2
VIII. Ignitor

FSYS Requirements

1. Ragnarok FSYS shall supply mass flow rates and pressure needed to injector, with at least 5% margin
2. Ragnarok FSYS lines/tanks shall have a MAWP above or at 720 psia
3. Ragnarok FSYS shall be reusable and maintainable
4. Ragnarok FSYS shall fail to a safe state
5. Ragnarok FSYS shall be able to refill or “top off” LOx
6. Ragnarok FSYS cryo-valves shall remain operable “indefinitely”

Ragnarok P&ID

1. Pressure fed system from 2ksi GN2 bottles, regulated to 700psia
2. Rented LOx dewar can be set to build pressure to allow for a liquid column to fill LOx tank from bottom
3. Kerosene will be pumped up vertically to fill tank from bottom
4. Torch GOx fed from LOx line through a “vaporizer”
5. Torch Kerosene fed from Kerosene line



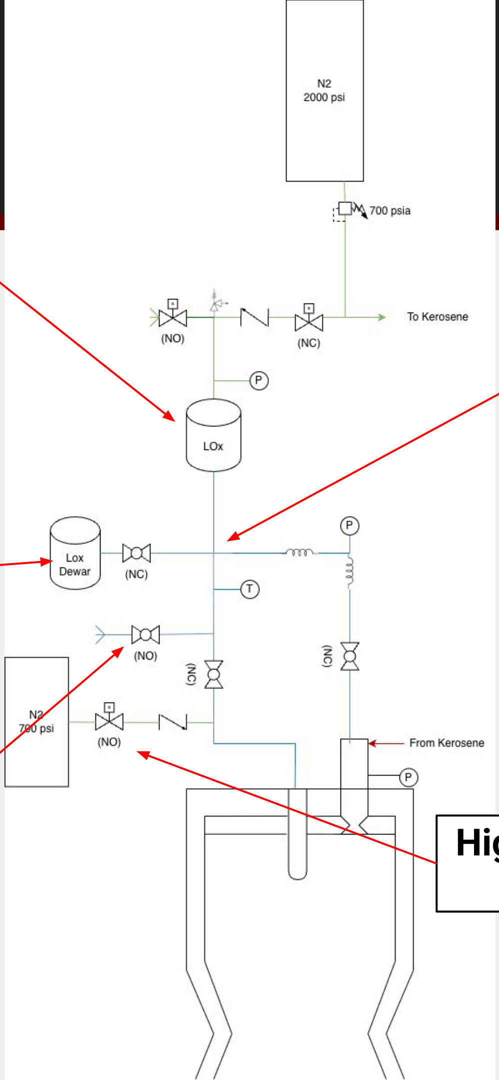
LOx P&ID

Tank MAWP >720psi, able to hold >=10s burn worth of prop

Tank level to be found via load cells and/or dP ducer

Dewar to hold enough prop for several fires, able to be set for pressure build enough for column up to top of test stand

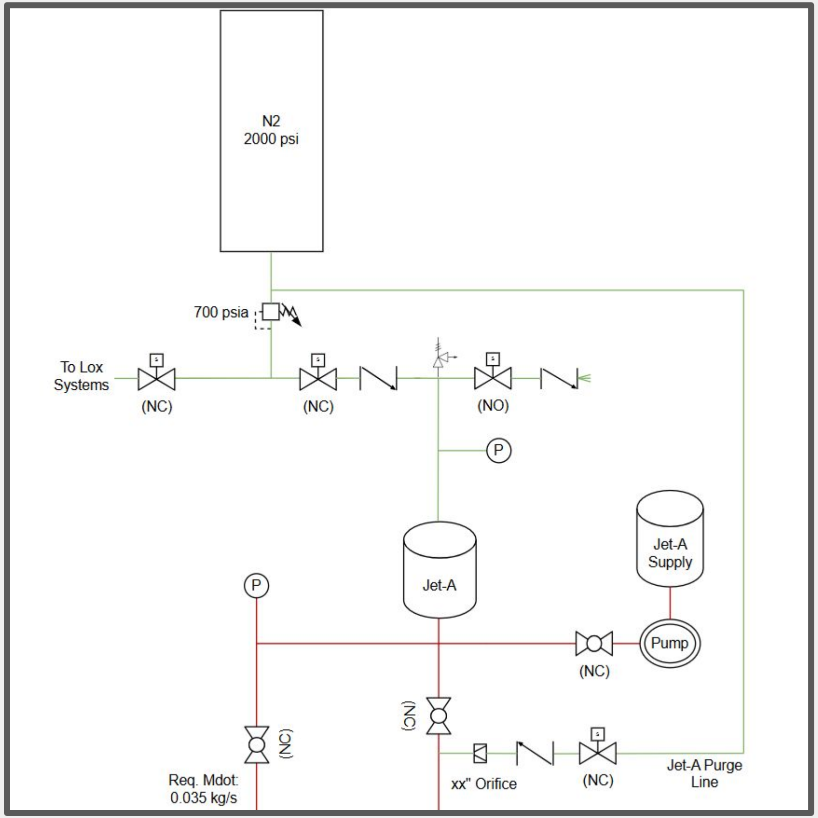
Fail open dump valve also allows flow path for farm chill LOx to chill in whole system

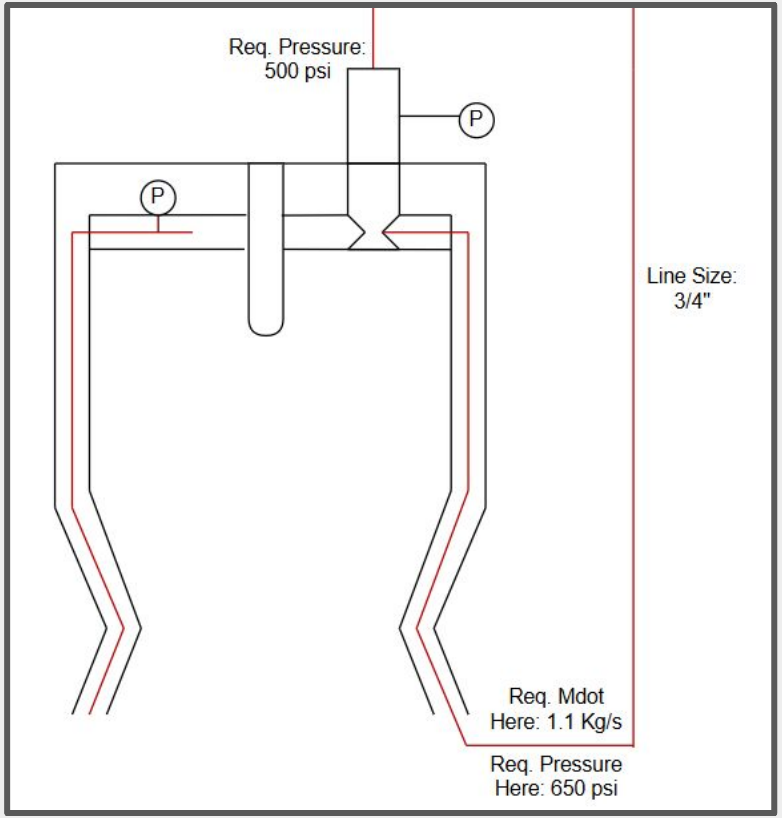


Decided NO tank iso valve here, whole LOx system will be chilled in up to main valve once prop load starts

High flow fail open purge for LOx and TCA

Jet-A P&ID

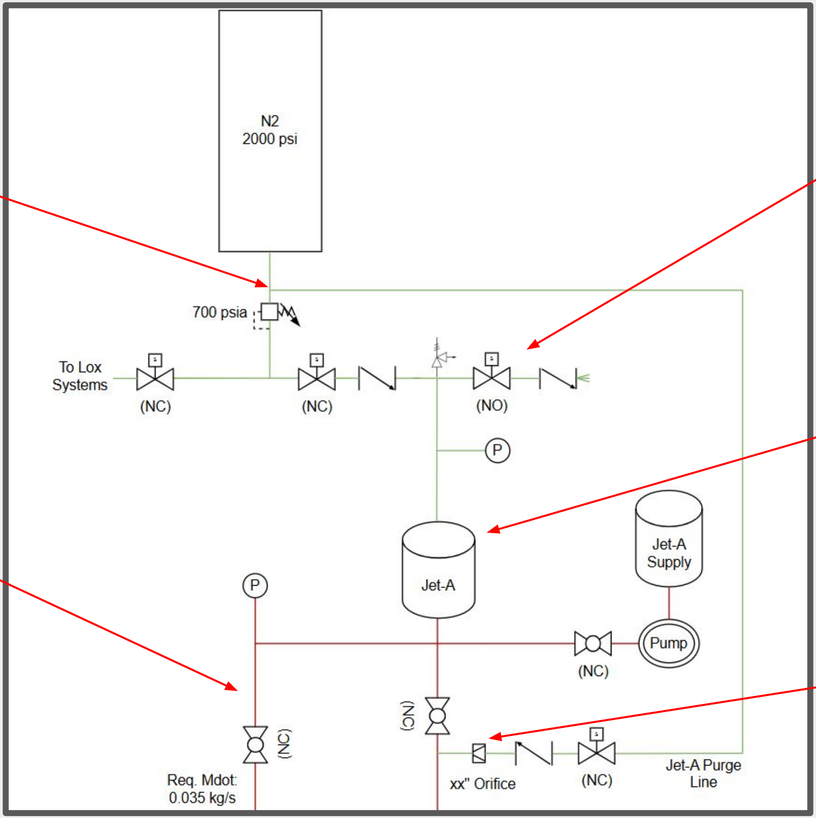

N2
2000 psi
700 psia
To Lox
Systems
(NC)
(NC)
(NO)
P
Jet-A
Supply
Jet-A
Pump
(NC)
P
(NO)
(NC)
Req. Mdot:
0.035 kg/s
xx" Orifice
(NC)
Jet-A Purge
Line


Req. Pressure:
500 psi
P
P
Line Size:
3/4"
Req. Mdot
Here: 1.1 Kg/s
Req. Pressure
Here: 650 psi

Jet-A P&ID

Purge GN2 line pre Reg

⅜” Line for torch ignitor

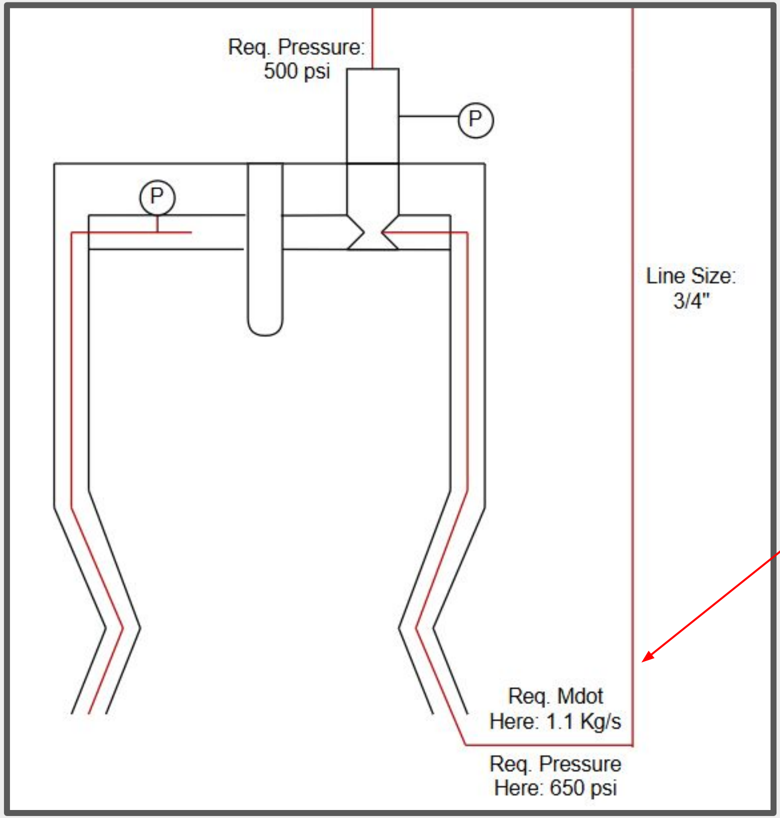


Fail open to vent GN2

Fuel dump tank by pressing down, & back through pump into fill tank.

Orifice to purge engine

Jet-A P&ID



Can likely get away with only 1 bend in the ¾" line

Jet-A Purge Orifice Calculator

Variables		Units
Gamma (Ratio of Specific Heats):	1.4	-
Gas Constant:	296.8	R (J/kg*K)
Gas Temperature	300	K
Pressure Before Orifice:	2000	psi
Pressure After Orifice:	500	psi
Pressure Ambient:	14.7	psi
Purge Orifice Discharge Coefficient:	0.8	-
Vent Orifice Discharge Coefficient:	0.8	-
Fuel Annulus Area:	0.00001	m^2
Pressure Before Orifice:	13890872.97	Pa
Pressure After Orifice:	3548732.972	Pa
Pressure Ambient:	101352.972	Pa
Critical Pressure Ratio:	0.899106091	-
Pressure After/ Pressure Before:	0.25	-
Flow Condition:	CHOKED	-
Choked Flow Mass Flux Constant K	0.00229470944	-
Choked Mass Flow (Vent):	0.06514648867	kg/s
Orifice Area:	**0.00000255472**	m^2
Orifice Diameter:	**0.07100570333**	in

$$\left(\frac{P^*}{P_0}\right) = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}}$$

$$\frac{P_2}{P_1} \leq \left(\frac{P^*}{P_0}\right)$$

$$K = \sqrt{\frac{\gamma}{RT_0}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}}$$

$$\dot{m} = A_t p_1 \left\{ \frac{\gamma}{RT_1}\left(\frac{2}{\gamma+1}\right)^{(\gamma+1)/(\gamma-1)} \right\}^{1/2}$$

$\dot{m}$ = critical mass flow

At = nozzle throat area,

p1 = inlet pressure,

R = universal gas constant,

T1 = inlet gas temperature and

γ is a physical property of the gas.

$$\dot{m}_2 = C_{d2} A_2 P_2 K$$

$$\dot{m}_1 = \dot{m}_2$$

$$C_{d1} A_1 P_1 K = C_{d2} A_2 P_2 K$$

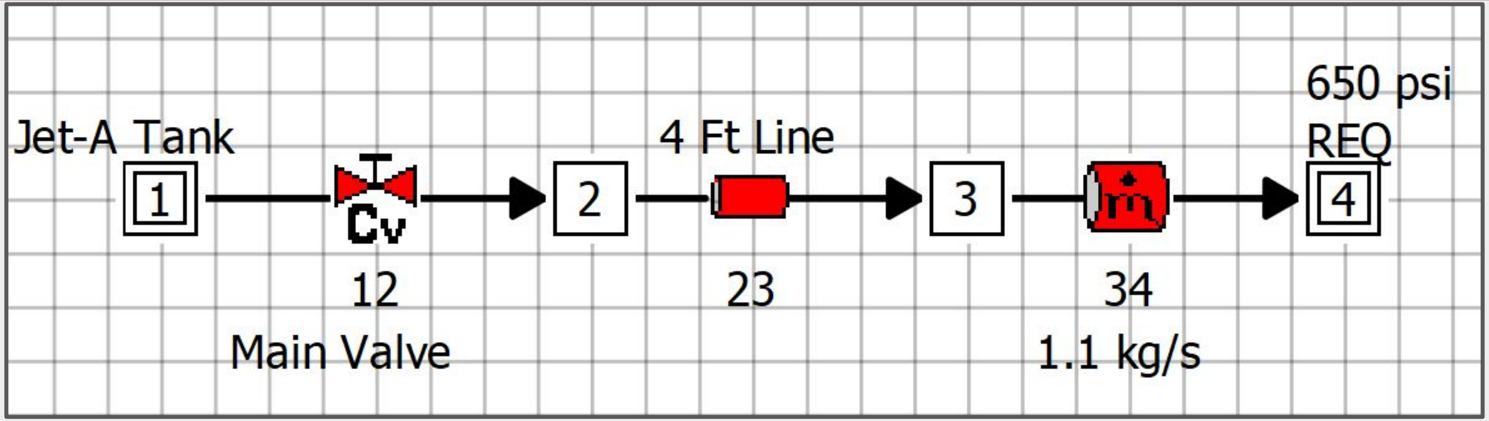
$$A_1 = \frac{C_{d2}}{C_{d1}} \frac{P_2}{P_1} A_2$$

Need orifice to take high pressure GN2 (2000 psi) to roughly 500 psi to purge engine.

Orifice will choke flow, and use the same mass flow rate as the max mass flow rate through the fuel annulus at 500 psi.

Uses fuel annulus Area to compute calculation.

Jet-A ¾” Line Rationale

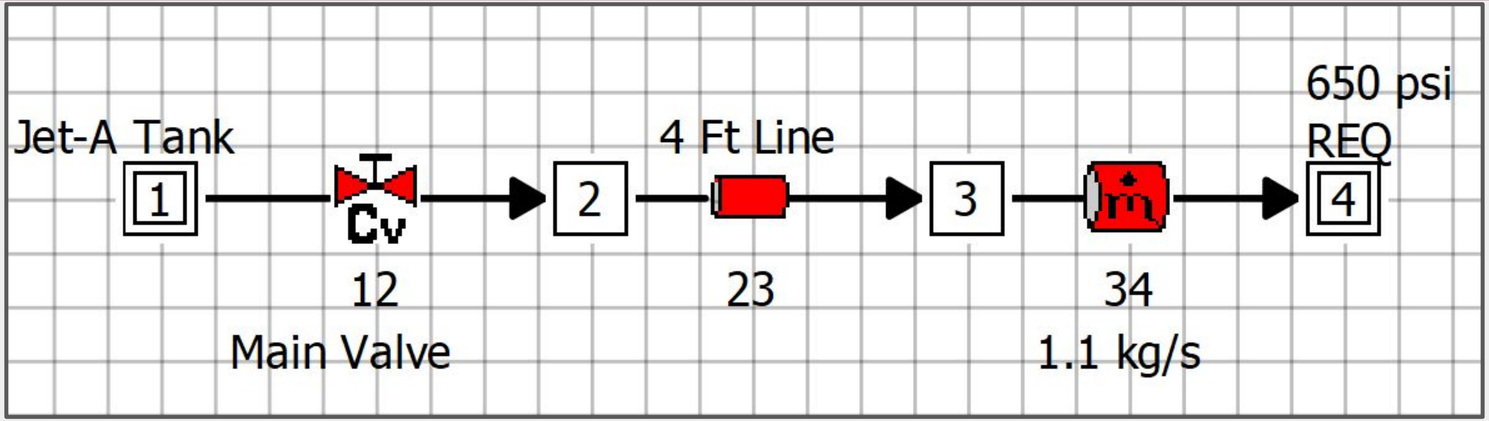


BRANCHES

BRANCH	KF (LBF-S^2/(LBM-FT)^2)	DELP (PSI)	FLOW RATE (LBM/S)	VELOCITY (FT/S)	REYN. NO.	MACH NO.	ENT. GEN. (BTU/R-S)	LOST WORK (LBF-FT/S)
12	1.7381E+01	7.0983E-01	2.4251E+00	6.4541E+01	8.2682E+04	1.6689E-01	1.1667E-05	4.9021E+00
23	3.9456E+03	1.6119E+02	2.4251E+00	6.4244E+01	8.2568E+04	1.6611E-01	2.6485E-03	1.1130E+03
34	0.0000E+00	-1.6190E+02	2.4251E+00	6.4758E+01	8.4861E+04	1.6723E-01	0.0000E+00	0.0000E+00

With ½ ” piping the pressure drop through the system is ~162 psi. This is sub-optimal :/

Jet-A ¾” Line Rationale

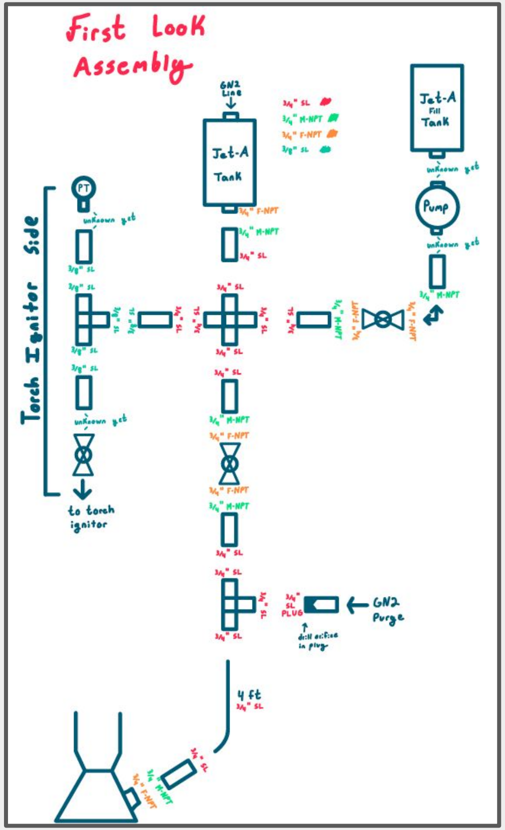


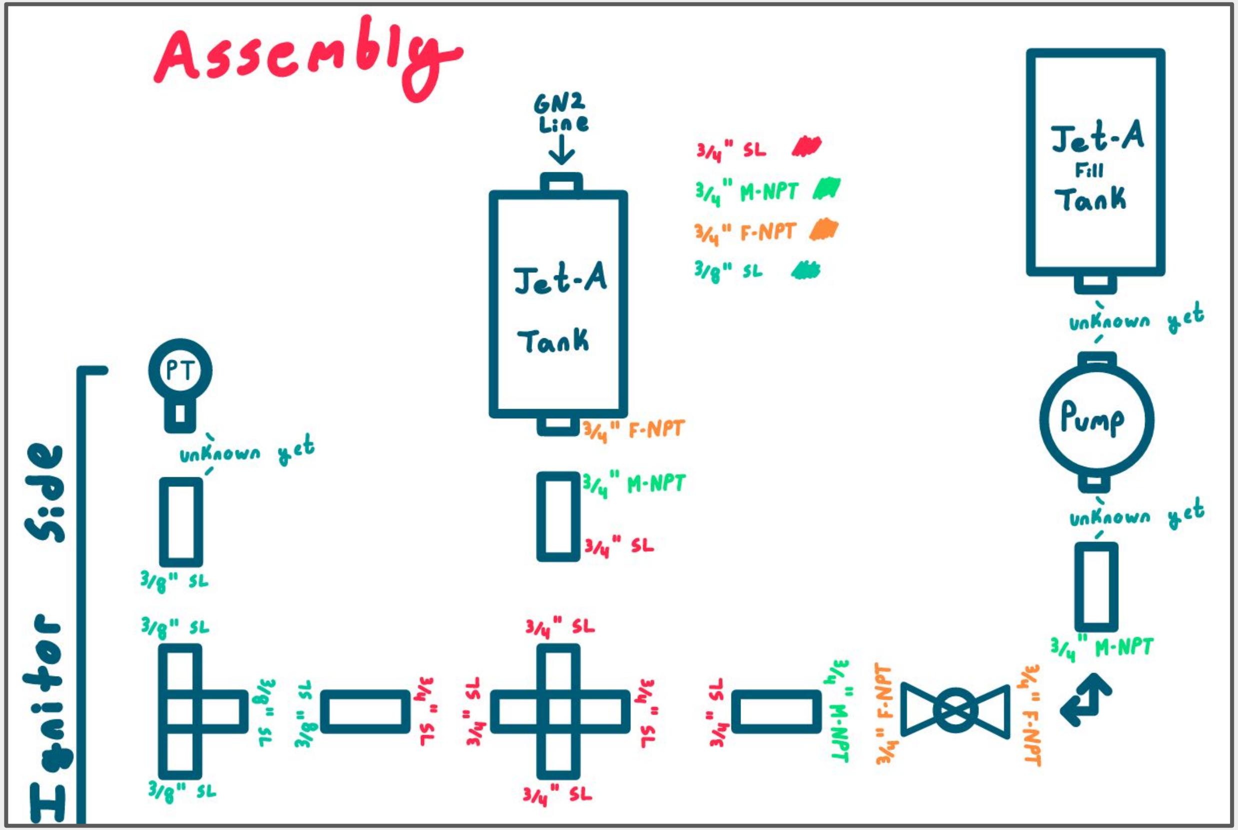
BRANCHES

BRANCH	KF (LBF-S^2/(LBM-FT)^2)	DELP (PSI)	FLOW RATE (LBM/S)	VELOCITY (FT/S)	REYN. NO.	MACH NO.	ENT. GEN. (BTU/R-S)	LOST WORK (LBF-FT/S)
12	3.4003E+00	1.3887E-01	2.4251E+00	1.9025E+01	4.4890E+04	4.9194E-02	2.2825E-06	9.5902E-01
23	1.5148E+02	6.1884E+00	2.4251E+00	1.9020E+01	4.4923E+04	4.9178E-02	1.0168E-04	4.2730E+01
34	0.0000E+00	-6.3272E+00	2.4251E+00	1.9049E+01	4.4976E+04	4.9251E-02	0.0000E+00	0.0000E+00

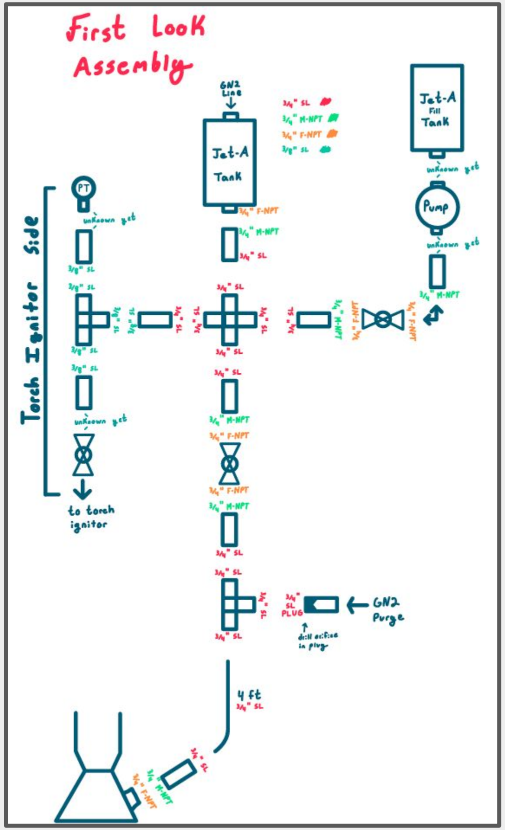
With ¾” piping the pressure drop through the system is ~6.3 psi. This is optimal!

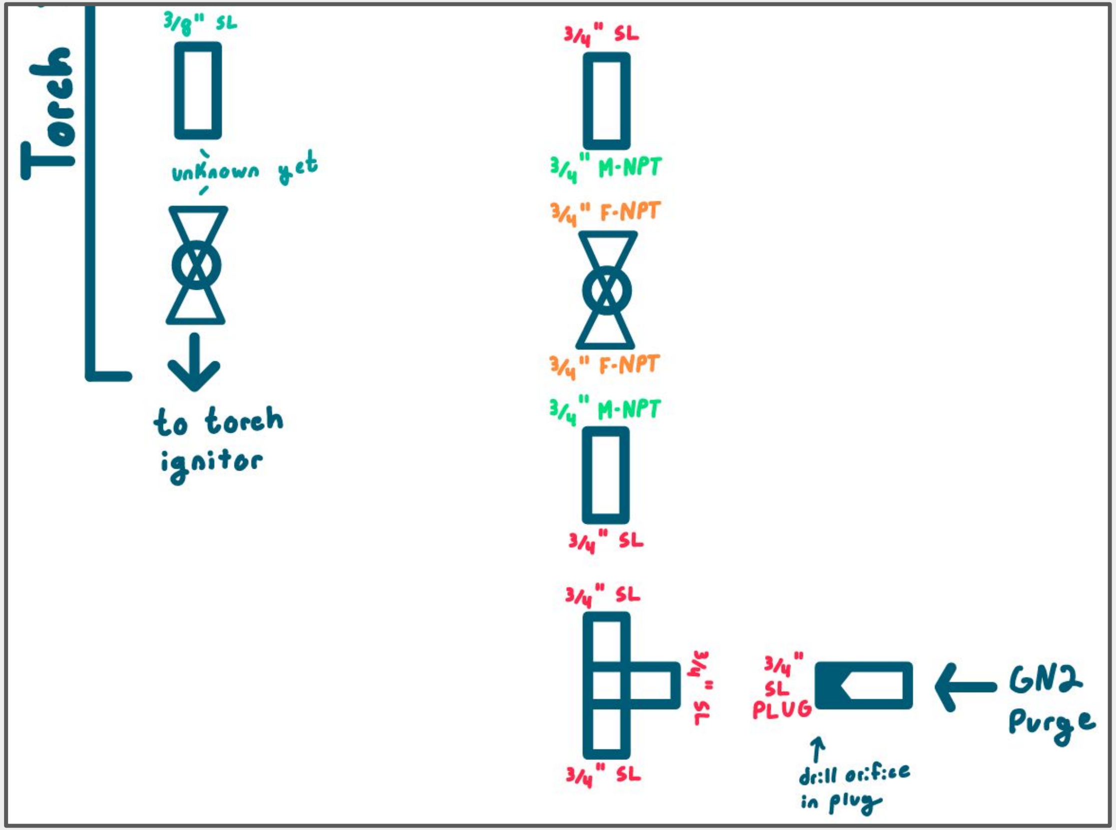
Jet-A Assembly Diagram



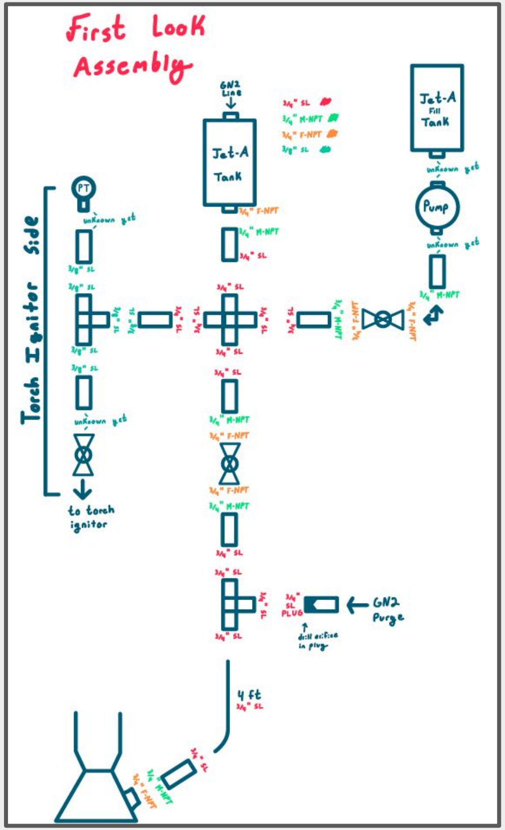


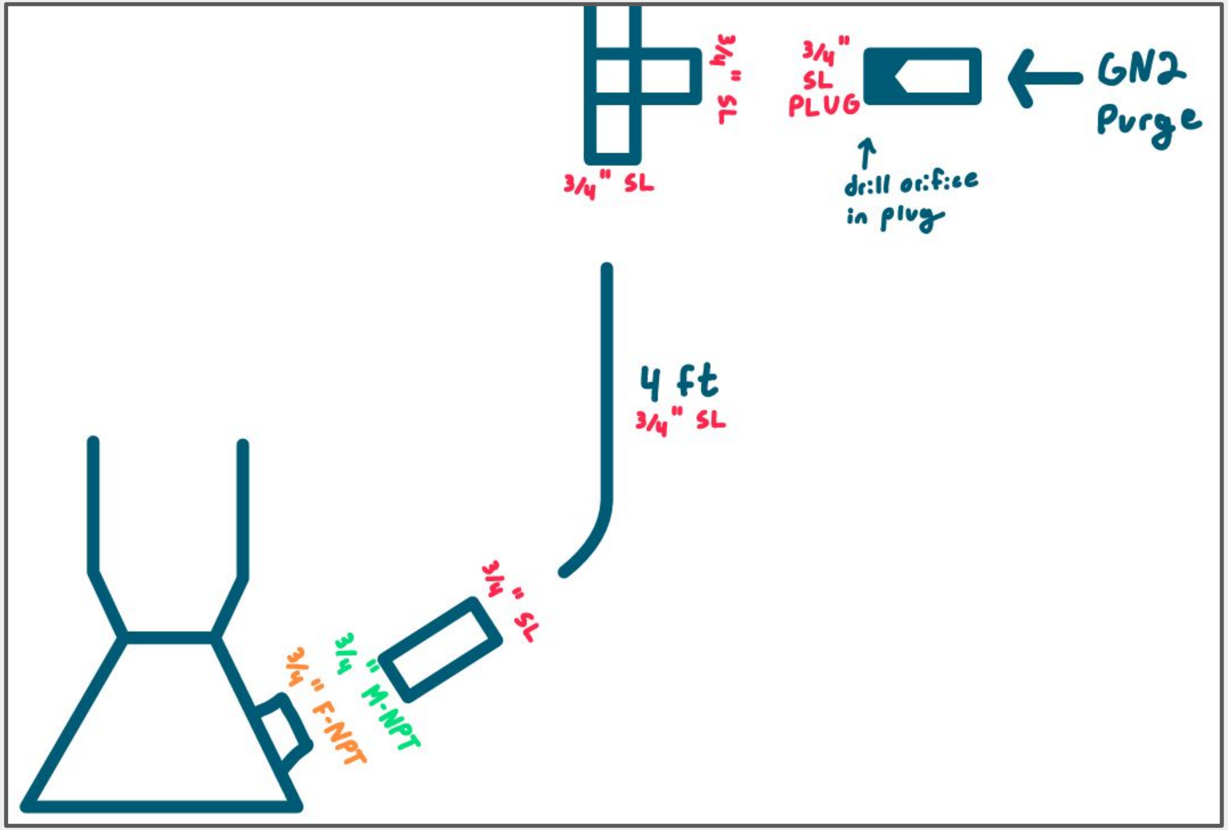
Jet-A Assembly Diagram





Jet-A Assembly Diagram





CONOPS

Assumed Chronological

1. Purge
2. Kerosene Prop Load
    a. Filled via pump from reservoir on ground to tank on top of test stand
3. LOx Prop Load and Chill
    a. No tank iso valve means full system chills in
4. Pressurize Tanks
    a. Press leak check
5. Torch Ignition
    a. Torch purge for fresh GOx, line purge for inert chamber
6. “GO”/Main Valves Open

Other OPs

- **Hold**
- **Refill**
- **Abort**
    - **Abort will dump all LOx**

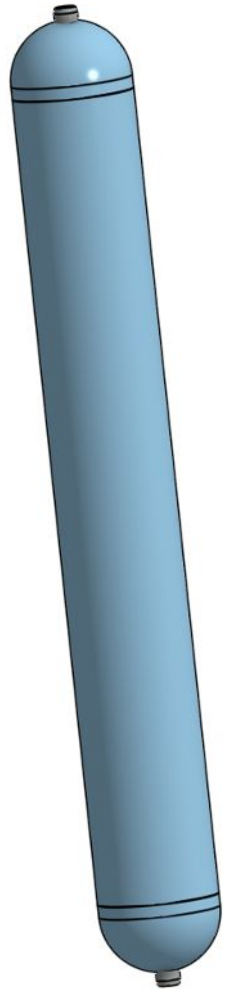
Tanks

Pipe Working Pressure Calculation		
Variable Description	Input	Units
Outside Diameter (OD) =	6.625	inches
Wall Thickness (T) =	0.134	inches
Allowable Stress (S) =	85000	psi
Safety Factor (SF) =??	3.00	See Above
Results		
Working Pressure (p)	1146.164	psi
Bursting Pressure	3438.491	psi

Pipe Working Pressure Calculation		
Variable Description	Input	Units
Outside Diameter (OD) =	6.625	inches
Wall Thickness (T) =	0.134	inches
Allowable Stress (S) =	34000	psi
Safety Factor (SF) =??	1.75	See Above
Results		
Working Pressure (p)	785.941	psi
Bursting Pressure	1375.396	psi

T4610	6" SCH 10 (6.625" OD X .134" wall X 6.357" ID) 304 Stainless Steel Pipe - Welded	9.30 lb/ft	4 Ft.	1	$480.24 ea.	Add To Cart

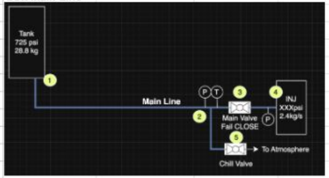
- Internal volume: 7.16 Gal, 27.1 L
- Max LOx mass: 69 lbs, 30.9 kg
- Max Fuel mass: 48.8lbs, 22.2 kg
- Height with domes: 4.5’
- Inside Diameter: 6.35”



LOx Pressure Ladder

Pressure @ chamber is 630 psia, 180 psia over chamber press target (450)

LOx Main Line



Inital Conditions		Units
Comodity	oxygen	
Temp	90	K
Pressure	700	PSIA
Target mdot	2.4	kg/s

Change these parameters to hit the desired values at the end, this was designed to be iterated quickly (even though the API be slow as balls sometimes)

Node 1.1	2ft Hose	
Commodity	oxygen	
Temperature	90	K
Mdot	2.4	kg/s
Abs Roughness	0.000015	m
ID	0.652	inch
Length	4	ft
Pressure	**700**	**psia**
Pressure SI	4826330.105	Pa_a
ID	0.0165608	m
Area	0.00021540337€	m2
Velosity	9.669610064	m/s
Density	1152.258113	kg/m3
Dynamic Viscosi	0.00020498511ε	Pa*s
Re	900155.8649	
f	0.01963001801	
Length	1.2192	m
Friction dP	77848.82121	Pa
Friction dP	11.29101691	psi
New Pressure	688.7089831	psia

Node 1.2	Panel Tube	
Commodity	oxygen	
Temperature	90	K
Mdot	2.4	kg/s
Abs Roughness	0.000015	m
ID	0.444	inch
Length	2	ft
Pressure	**688.7089831**	**psia**
Pressure SI	4748481.284	Pa_a
ID	0.0112776	m
Area	0.00009989028ξ	m2
Velosity	20.85446742	m/s
Density	1152.096569	kg/m3
Dynamic Viscosi	0.000204832111	Pa*s
Re	1322837.909	
f	0.02133237205	
Length	0.6096	m
Friction dP	288884.683	Pa
Friction dP	41.89918089	psi
New Pressure	646.8098022	psia

[illegible]	[illegible]	[illegible]
Mdot	2.4	kg/s
Pressure	**646.8098022**	**psia**
ID	0.444	in
# 90deg (short radius)	0	
# 90deg (long radius)	0	Assuming 1/2" NPS, update K value in "SUM K" row
# Tee (straight)	1	
# Tee (branch)	0	
Sum other K factors	0	
K	0.54	
ID	0.0112776	m
Area	0.00009989028ξ	m2
Pressure SI	4459596.601	Pa
Density	1151.495536	kg/m3
Velocity	20.86535261	m/s
g	9.81	m/s2
dP bends	11.98246622	dPa
dP bends	0.00173790979⁄	psi
New Pressure	646.8080643	psia

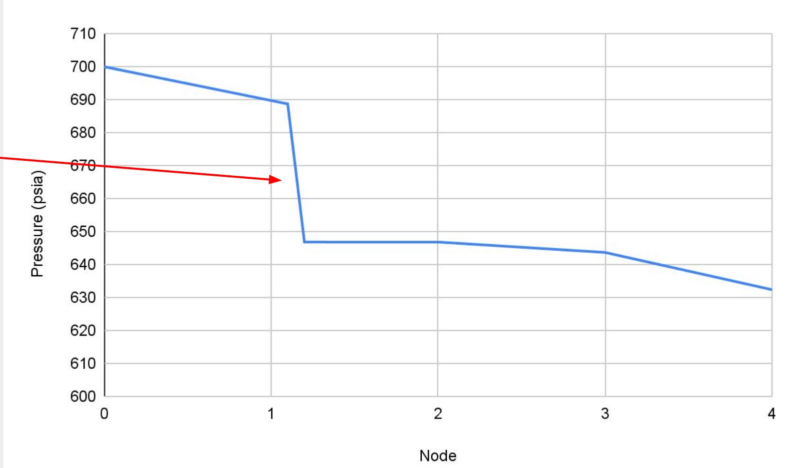
[illegible]	[illegible]	[illegible]
Mdot	2.4	kg/s
Pressure	**646.8080643**	**psia**
CdA	0.526	in2
CdA	0.00033935416	m2
Pressure SI	4459584.619	Pa
Density	1151.495511	kg/m3
CdA dP	21718.20279	Pa
CdA dP	3.149959	psi
New Pressure	643.6581053	psia

e 4	2ft Hose	
mmodity	oxygen	
perature	90	K
Mdot	2.4	kg/s
Pressure	**643.6581053**	**psia**
Abs Roughness	0.000015	m
ID	0.652	inch
Length	4	ft
Pressure SI	4437866.416	Pa
ID	0.0165608	m
Area	0.00021540337€	m2
Velosity	9.67639452	m/s
Density	1151.450225	kg/m3
Dynamic Viscosi	0.00020422136ς	Pa*s
Re	903522.2764	
f	0.01962852419	
Length	1.2192	m
Friction dP	77897.51364	Pa
Friction dP	11.29807915	psi
New Pressure	632.3600261	psi

Assuming a starting tank press of 2.4kg/s and 700 psia starting (head losses neglected)

Basically all losses through 2ft of ½" tube on main panel (42psid)

Using ¾" hoses for less dp through tank to panel and panel to chamber hoses



LOx Valve Mod

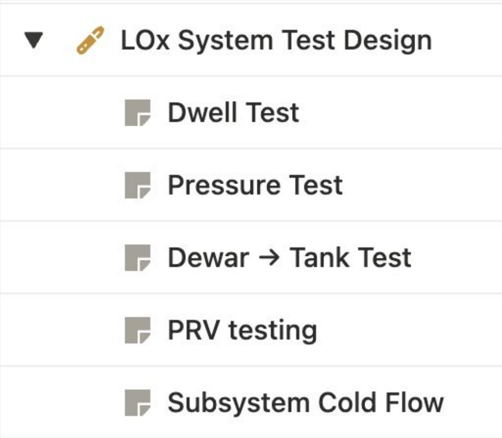
Doc summarizing work: Link

TLDR: We can reliably use a $60 ½" ball valve (PTFE seals) with a weep-hole drilled in the ball for cryo applications if we apply heaters to the stem. We know this to be true for 60W (2x 24V heaters), but are doing ongoing testing with LN2 to see how low we can get the power and make sure this is *very* reliable

Tube

LOx Tests

- Dwell test - sit with LN2 for a few hours and check the valve works every ~15 mins
- Pressure Test - pressurize the LN2 and ensure it clicks
- Dewar -> Tank - practice fill ops
- PRV Testing - ensure PRVs work as expected and while chilled
- Subsystem Cold Flow - Run whole thing babyyyyyy



LOx Path Forward

- PDR
  - Show math for
    - Pressure Ladder (pressures and system mdot)
    - Tanks
    - GN2 flow rates
    - Dewar capabilities
  - Show all components
    - Valves
    - Tank stuff
    - Hoses
  - Show CAD for
    - Whole system
  - Show detailed CONOPs along side the P&ID
  - Outline tests/path forward after PDR
    - Show how this helps us write a CDR

GN2 Consumption Expectations

States	Mass Consumption (kg)	Mass Flow Rate (kg/s)
LOx Purge	3.45	0.345
Kerosene Purge	1.12	[redacted]
Pressurization	1.14	[redacted]
Fire	2.6	0.26
Post Fire Purge	3.45	0.69
Total Consumption	11.76 kg	[redacted]

Using 3 Nitrogen bottles we expect to have access to 20 kilograms of nitrogen.

Calculating the nitrogen consumption at each stage we expect to use roughly 55% of our nitrogen in total.

Pneumatics

All cryogenic ball valves shall be pneumatically controlled

Current Pneumatic Options include

- Pre assembled Ball valve and Pneumatic Actuator
- Buying Seperate Ball Valve and Pneumatic Actuator

Valves will be powered through compressed air

- We will test our current air compressor to determine how long we the system can be powered with our air compressor.

Torch Ignitor - LOX Tapoff

- Vaporizer line will be tapped off from LOX supply line to ensure GOX delivery to torch ignitor subsystem
- Will consist of ⅛" coiled SL tubing to feed into ignitor
- Heat transfer from ambient air to vaporizer line will assist in natural LOX -> GOX boiling
    - May require direct line heating: cartridge heaters, resistive heating tape?
- Vaporizer calculator accounts for 1D heat transfer model of LOX -> GOX under expected flow conditions
    - Can be used to iteratively solve for necessary line length as well as necessary applied heat flux for liquid to gas phase change

Inputs
LOX Mass Flow [kg/s]
Operating Pressure [Pa]
LOX Inlet Temperature [K]
Desired GOX Outlet Temperature [K]
Tube Inner Diameter [m]
Tube Outer Diameter [m]
Tube Material Thermal Conductivity [W/m-K]
Total Coil Length [m]
Hot Side Temp [K] (Average of Inlet & Outlet)
External heat transfer coefficient (air) [W/m^2-K]

AVI

I. Requirements
II. GIGAPCB Compatability
III. Valve Heaters
IV. Spark Plug Actuation

AVI Requirements

- Ragnarok AVI shall use GIGA PCB
- Ragnorak AVI shall sufficiently power all valve heaters
- Ragnorak AVI shall control spark plug actuation

GIGA PCB Compatibility

GIGA PCB is capable of handling

- 12 thermocouples
- 12 ~ 16 relays (upgradable)
- 6 load cells
- 12 pressure transducers

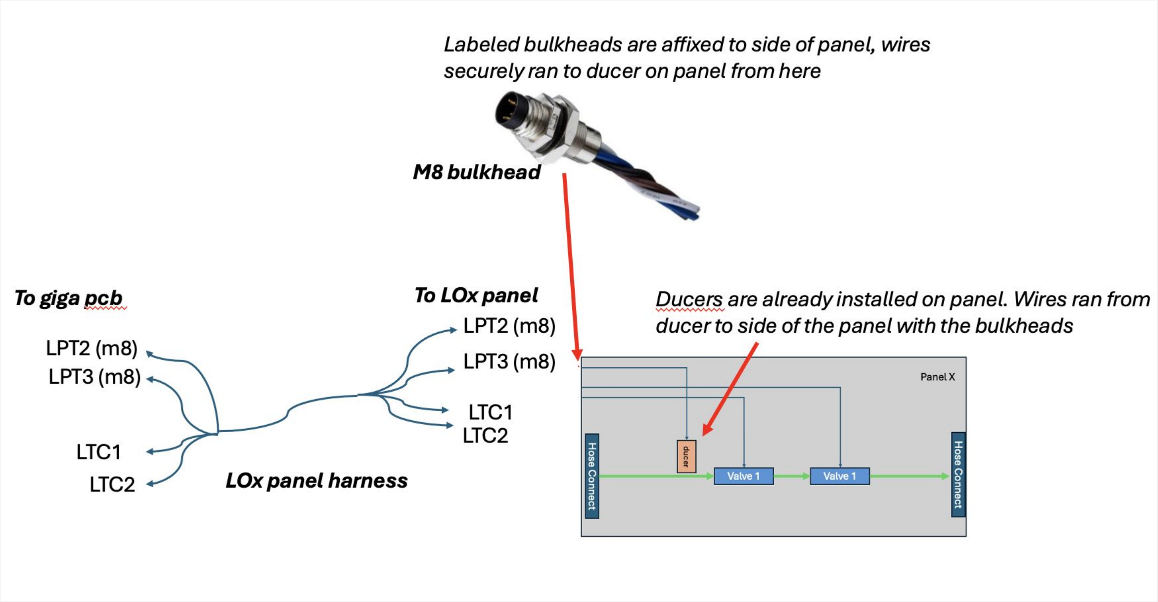
Ragnarok will use

- No more than 12 termocouples
- 8 ball valves + 5 solenoids + spark plug (total of 14 relays)
  - Will upgrade the GIGAPCB to 16 relays (add more relays to the board or add a separate board)
- Elysium 2 used 5 LCs, same for Ragnarok
- No more than 12 pressure transducers (currently at least 6)

Panels

Make a harness for each panel, made from wires taped together

- Adds localization, which helps with organization and troubleshooting.
- Use tape every 6”, label each harness and wires with ID



Valve Heaters

Top: 12V 12W, Middle: 12V, 7W, Bottom: 24V, 30W

Planned to be powered by a generator due to high power draw (4 heaters, total max of 240 W)

- 1000 Wh battery (around $400) can last 4 hours
- 1000 W generator (around $150) can last 10+ hours per gallon

Tests were conducted using the bottom heater

- Valve was hot, LN2 introduced while closed
  - Valve remained sealed, opened well
- Valved chilled in, steady stated cold
  - Valve actuated well
- Heaters turned off, let stem freeze over
  - Valve opened well, not sure if fully closed

Conclusion: Should turn the valve heaters on the moment we introduce cryogenic fluid, and keep them on while there still is cryo

STR

I. Requirements
II. E2 Test Stand Overview
III. Test Stand Upgrades
IV. Tank Support Structure (wood)

STR Requirements

- Ragnarok shall use Elysium 2.0 test stand
- Ragnarok shall have a new tank support structure
- Ragnarok shall have the tank support structure mounted above the Elysium 2.0 test stand
- Ragnarok shall have a new interface plate for connection between the Thrust Chamber and the test stand
- Ragnarok shall have improve the Elysium 2.0 test stand stiffness and safety factor to withstand 2000 lbf with a FoS 4

E2 Test Stand Overview

- Ragnarok reuses the Elysium 2.0 test stand originally designed for 1500 lbf thrust with a FoS of 4
- Ragnarok is rated for 2000 lbf, requiring an increase in effective load capacity to maintain a FoS of 4
- Primary structural limitation is the bolted connection, not the stand frame, governed by shear and joint compliance
- Brackets between I beam and mounting plate will potentially be welded to the I beam to improve stiffness and load transfer, reducing reliance on bolt shear alone

E2 Test Stand Overview

- Ragnarok will use a new designed interface plate to connect the Thrust Chamber to the existing mounting plate
- Ragnarok will use a new Tank Support Structure (wood) to support 2 propellant tanks, 4ft tall & 6in diameter
- Above mentioned upgrades will enable the reuse of Elysium 2.0 test stand while safely supporting higher thrust loads and new engine tank configiration

SAFE

I. Requirements
II. CONOPS
III. FOD mitigation/Cleaning
IV. Site Safety
V. Venting/Purging
VI. New Hazards

Requirements

- Ragnarok shall go to a fail safe state in the event of power loss or anomaly
- The test site shall minimize the amount of flammable material present near LOx hardware
- Operating personnel shall wear appropriate PPE while handling propellants
- Propellants will be transported according to D.O.T standards

CONOPS

- T-Days - FSYS Lines are cleaned, dried and capped/bagged and prepped for testing & transport after Dry Runs/Assemblies
- T-Minus hours - SAFE/Assembly team preps site, assembles stand and slowly connects non LOX FSYS lines. Safety criteria for each team checked.
- T-Minus 1 hour - LOX arrives and gets placed at storage point on site. Lines are connected. Access to test stand heavily controlled
- T-Minus minutes - Prop Load and final go/no go
- T-Minus seconds - automated ignition sequence starts

FOD Mitigation/Cleaning

- Due to the flammability of LOx and GOx, extra precaution must be taken for any FSYS hardware that LOX will flow through.
- Minor particulates and oil that were a non issue in Elysium class engines now present major system risk in LOX lines.
- Cleaning will be done according to ASTM Man. 36 and other resources. Hydrocarbon based oils & particulates are to be thoroughly removed.
- Additional physical barriers like Precision Pipe Products covers and double bagging prior to transport will be looked into further as design becomes finalized
- Industry Standard methods will be researched further

Safety

- Control to test stand will be heavily controlled, clothing standards will be enforced on site.
- Design for Safety will be done from the beginning of the design process
- Site inspections will be conducted prior to setting up for any test involving LOx, marking out safety area and prep site for test stand and infrastructure
- Team meetings prior to fire will be used to brief everyone on dangers of LOX/GOx alongside hotfire/rehearsal operations and contingencies.
- Any member who has potential to approach test stand will be briefed in depth on increased hazards and appropriate PPE

Venting/Purging

- Oxygen filled lines when purged must be filled with a non reactive gas that also does not contaminate the FSYS.
- GN2 was used previously and shall continue to be used
- Due to ambient temperature being higher than state transition temperature of LOx, venting of GOx will need to become part of operating procedure to ensure fluid lines are saturated with LOx and not GOx
- Systems design will need to be taken into consideration on the location of the vent valve of GOx to prevent any event of saturation of GOx anywhere in the system

New Hazards

- Previous propellants were stable and not as difficult to transport, LOx increases complexity of transport
- LOx increases flammability of flammable materials if spilled onto it, therefore precaution must be taken to prevent any flammable materials in spill areas
- GOx is highly flammable, system design must minimize opportunities for short circuits to occur in areas where GOx could pool.
- In the event of a spill of fuel and LOX resulting in a mixture, explosion hazard is present until LOX evaporates.
  - Possibly look into fans or other events to blow any pooled GOx/LOx away